

Practice Quiz: Cognition / Deep Temporal Models

(Advanced Theory)

Name: __ Date: __

Part A: Multiple Choice

1. Deep temporal models address: A) Only spatial problems B) The challenge of inference across multiple timescales — using hierarchical levels that operate at progressively slower temporal scales C) Only auditory processing D) Static pattern recognition
2. In a deep temporal model, ascending messages carry: A) Motor commands B) Prediction errors from lower levels, summarizing what the model got wrong — informing higher-level context estimation C) Random noise D) Observations directly
3. Descending messages carry: A) Reward signals B) Top-down predictions from higher levels, constraining lower-level inference with contextual expectations C) Error signals D) Motor outputs
4. Multi-scale planning implements: A) Only immediate reactions B) Nested policy evaluation where higher-level policies constrain lower-level EFE calculations C) Random action selection D) Only habitual behavior
5. Counterfactual reasoning is implemented by: A) Ignoring alternatives B) Running the generative model forward under different clamped policy choices, comparing expected free energy outcomes C) Only remembering the past D) Waiting for outcomes
6. Higher levels evolve more slowly because: A) They are less important B) They represent more abstract, temporally stable features — contexts change less frequently than sensory details C) Higher neurons fire less D) There are fewer neurons
7. Imagination emerges from: A) Random brain activity B) Running the generative model in "generative mode" — sampling from the model without constraining observations C) External stimulation D) Memory replay only

Part B: Essay Questions

1. Design a 4-level deep temporal model for music composition. Specify: (a) what each level represents (timbre/notes/phrases/sections), (b) the transition dynamics at each level, (c) how ascending and descending messages flow during composition, (d) how the model generates novel music (creativity). (500 words)
2. How do deep temporal models relate to the concept of narrative self? The highest level of a deep temporal model represents the agent's life narrative — the slowest-changing self-relevant beliefs. Analyze: (a) how narrative identity emerges from temporal depth, (b) what happens when the narrative level is disrupted (trauma, psychosis, loss of meaning), (c) how psychotherapy might work by restructuring the highest-level priors. (400 words)
3. Compare deep temporal models with other hierarchical models: (a) hierarchical reinforcement learning (options framework), (b) predictive coding, and (c) recurrent neural networks with attention. What does each handle well? Where does the deep temporal model framework add value? (400 words)